

The Landscape Environment

A Ground-Up Formalization of Coarse-Grained Structure Learning with an Endogenous Observation Frontier

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Abstract

We specify, from first principles, a learning *environment*: a deep ground-truth world with rich hidden substructure, a population of communities that each begin from a different *coarse* model and must learn the world from scratch, and—the one ingredient that makes the environment more than a curriculum—an *observation frontier* that unlocks new channels as a function of the structure a community currently holds. We give the same object four readings, each borrowed from a different field: a Gibbs *energy landscape* (statistical physics), a *graphical model under structure learning* (probabilistic AI / active inference), a *coarse-graining* (renormalization and Markov lumpability), and an *accessible poset* (order theory). The last reading is the payoff: the unlock map is a monotone operator on a finite lattice of structures, its least fixed point is fixed by a community’s coarse seed, and the *multiplicity of fixed points across seeds is exactly structural pluralism*—a path-dependence that needs no tuned exploration term and that a social coupling graph can either preserve or collapse.

1 The environment in one object

The environment has three parts and one loop (Fig. 1). A *world* W fixes a ground-truth law over fine variables $x \in \mathbb{R}^n$ with n large and a deeply structured dependency graph. A population of *communities* $\{M_c\}_{c \in C}$ each hold a model that is, at the outset, far *coarser* than W and that they grow by learning. An *observation frontier* F returns, for any held structure G , the set of measurement channels currently available—and that set is itself a function of G . The loop closes: the channels a community has unlocked deposit information, the information grows the held structure, and the grown structure unlocks further channels. Everything below is this one object, read four ways; the readings agree on the mathematics and disagree only on which field’s intuition they borrow (§6 collects the dictionary).

2 The world as an energy landscape

Read the world thermodynamically. A configuration x has an *energy* $E(x)$ and the world’s law is the Gibbs measure $p(x) = Z^{-1}e^{-E(x)}$, $Z = \int e^{-E(x)} dx$ (Jaynes, 1957). For any strictly positive law the Hammersley–Clifford theorem makes the energy a sum of clique potentials of an undirected graph $\mathcal{G}^* = (V, \mathcal{E}^*)$,

$$E(x) = \sum_{C \in \text{cl}(\mathcal{G}^*)} \varphi_C(x_C), \tag{1}$$

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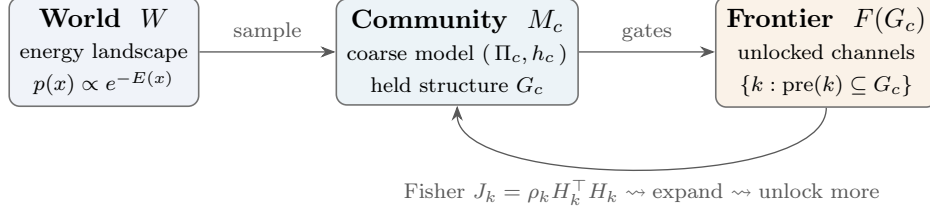


Figure 1: The environment as three objects and a feedback loop. The world is sampled through whatever channels the community has unlocked; the resulting information grows the held structure G_c ; the grown structure widens the frontier $F(G_c)$. §2–5 formalize each box and the loop.

so the *shape of the energy is the graph*: $x_i \perp x_j \mid x_{\setminus ij}$ exactly when no clique holds both (Besag, 1974). The tractable corner we use throughout is the Gaussian (Gauss–Markov) field, where the potentials are quadratic and

$$E(x) = \frac{1}{2} x^\top \Pi^* x - h^{*\top} x, \quad p(x) = \mathcal{N}((\Pi^*)^{-1} h^*, (\Pi^*)^{-1}), \quad (2)$$

with the *precision* $\Pi^* \succ 0$ playing the role of the energy’s Hessian; its zero pattern is $\mathcal{G}^*$. “Deep substructure” is then a literal statement about Π^* : order the fine variables into a dense, high-precision *core*, a layer of *mechanisms*, and a sparse observable *belt*, and Π^* is block-structured by depth (Fig. 2, right). Beyond the Gaussian corner the same energy carries the feature that makes the world hard: many *metastable basins* organised hierarchically—the ultrametric landscape of spin glasses (Mézard et al., 1987), the associative-memory energy of Hopfield (1982). A basin is a candidate theory; the barrier between two basins is the energy a learner must pay to move between them—the cost of revising a commitment (Fig. 2, left). The world is not a target to be guessed but a landscape to be traversed.

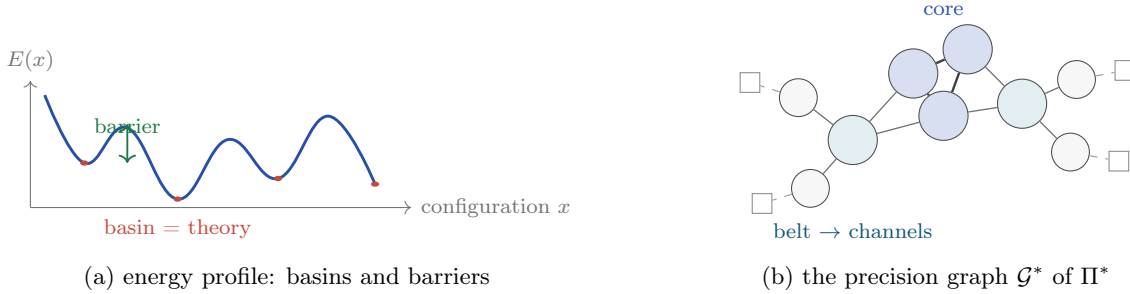


Figure 2: The world is an energy landscape. a In non-Gaussian energies the landscape is rugged: basins are candidate theories and the barrier between them is the revision cost. b The sparsity of the Gaussian precision Π^* is the dependency graph; depth runs core (dense) $\rightarrow$ mechanisms $\rightarrow$ belt, and only the belt touches observation channels (squares).

3 The world as a graphical model, and learning as structure search

Read the same world causally. Factor the Gaussian field as a linear structural equation model on a weighted DAG,

$$x = Ax + \zeta, \quad \zeta \sim \mathcal{N}(0, D^{-1}), \quad \Pi^* = (I - A)^\top D (I - A), \quad \mathsf{T} = (I - A)^{-1} = \sum_{k \geq 0} A^k, \quad (3)$$

so $x = \mathsf{T}\zeta$ and $p(x) = \prod_i p(x_i \mid x_{\text{pa}(i)})$ is the Markov factorization of Pearl (2009); Koller & Friedman (2009); T is the *propagation operator* that carries a perturbation at one node to all its descendants.

A community is a *belief* over this structure, held in the same information form, $M_c = (\Pi_c, h_c)$ on a node set $V_c \subseteq V$. The world enters a belief only through measurement. An *observation channel* k reads a linear combination $o_k = H_k x + \varepsilon_k$, $\varepsilon_k \sim \mathcal{N}(0, \rho_k^{-1} I)$, and the conjugate update adds its Fisher information to the precision,

$$\Pi_c \leftarrow \Pi_c + \sum_{k \in F(G_c)} J_k, \quad J_k = \rho_k H_k^\top H_k, \quad j_k = \rho_k H_k^\top o_k, \quad (4)$$

the sum running only over *unlocked* channels $F(G_c)$ —the coupling to §5. The Fisher term J_k is also the metric of the statistical manifold on which the belief moves, so learning is natural-gradient descent on variational free energy (Amari, 2016).

Learning the *structure*, not just the state, is two moves of asymmetric cost (Fig. 3). Each candidate edge e carries an automatic-relevance-determination prior precision α_e , with $\alpha_e \rightarrow \infty$ pinning the edge off and $\alpha_e \rightarrow 0$ freeing it (MacKay, 1992; Neal, 1996; Tipping, 2001). *Reduction* prunes an edge the data no longer support; because a reduced model shares its likelihood and differs only in its prior, the change in evidence is closed form—the Savage–Dickey ratio of Bayesian model reduction,

$$\frac{p(o | \tilde{M})}{p(o | M)} = \mathbb{E}_{p(\theta|o)} \left[\frac{\tilde{p}(\theta)}{p(\theta)} \right], \quad \Delta F > 0 \Rightarrow \text{prune}, \quad (5)$$

read off the already-inverted posterior (Friston et al., 2018). *Expansion* proposes a commitment not yet represented—bordering a new node N wired at free priors and letting reduction carve it back (Friston et al., 2017). Expansion is the costly half: a genuinely new node may open an observable channel and so enlarge the likelihood’s support, the one real inversion the cheap closed form is spared. This is the formal seam at which §5 cuts—the move that *unlocks an observation* is exactly model expansion.

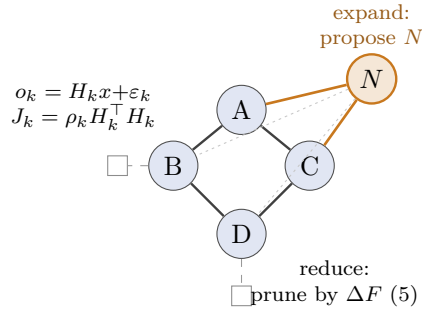


Figure 3: Learning as structure search. Observation channels (squares) deposit Fisher information (4) into the belief; *expansion* borders a new node N at free priors (amber) and *reduction* prunes the candidate edges the data do not hold (faint) in closed form (5). Expansion is the move that can open a new channel—the hinge to §5.

4 Coarse-graining: where a community starts

A community does not start with $\mathcal{G}^*$; it starts *coarse*. Formalize a coarse model as a partition $P_c = \{B_1, \dots, B_m\}$ of the fine nodes V into super-nodes, with aggregation map $S_c \in \{0, 1\}^{m \times n}$, $(S_c)_{ai} = 1 \Leftrightarrow i \in B_a$. The coarse variable is $y = S_c x$, and integrating out the within-block fluctuation gives the effective (Gaussian) energy

$$e^{-E_c(y)} \propto \int e^{-E(x)} \delta(S_c x - y) dx \quad \Rightarrow \quad \Pi_c = (S_c (\Pi^*)^{-1} S_c^\top)^{-1}, \quad (6)$$

the block-aggregated precision (the Schur reduction onto the super-nodes; in the repository this is `bmr.schur_marginalize`). Equation (6) is one blocking step of the renormalization group: replace fluctuating fine detail by a super-node and read the induced effective couplings (Kadanoff, 1966; Wilson, 1975; Jona-Lasinio, 1975). The coarse model is *exact*—a faithful lower-resolution copy—only when the partition is strongly lumpable, i.e. respects the dependency structure so that block members are interchangeable (Kemeny & Snell, 1976); generic partitions are not lumpable, and then a coarse model is strictly weaker: it senses only aggregates, resolves the world more slowly the coarser it is, and under forgetting plateaus at a residual it can never clear. Crucially, **refinement is expansion**: splitting a super-node into two is the addition of a node (§3), a single step *up* the partition lattice (Fig. 4). The lattice of partitions, ordered by refinement, is therefore the scaffold on which the next section’s frontier climbs—and where a community sits on it at $t = 0$ is its *seed*.

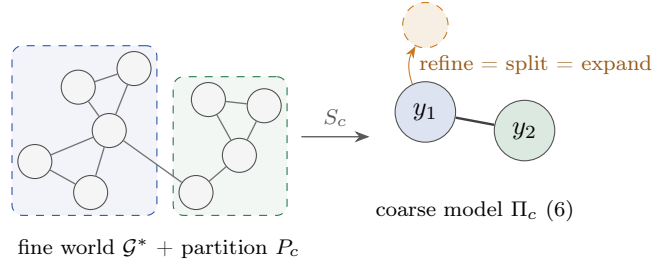


Figure 4: Coarse-graining as a starting structure. A partition P_c of the fine world (left, two blocks) aggregates by S_c to a coarse model Π_c (right) via the renormalization step (6). Refining—splitting a super-node—is a model-expansion move, one step up the partition lattice. A community’s seed is where it sits on that lattice at $t = 0$.

5 The endogenous observation frontier

Here is the ingredient that makes the environment its own thing. Let Ω be the ground set of structural elements (candidate nodes and edges), and let a community’s *held structure* $G \subseteq \Omega$ be the subgraph it currently commits to; its seed $G_c(0)$ is the coarse model of §4. Each channel k carries a *prerequisite* $\text{pre}(k) \subseteq \Omega$, and the world is legible to a community only through the channels its structure has earned.

Definition 1 (Frontier). The *unlocked frontier* at held structure G is $F(G) = \{k \in K : \text{pre}(k) \subseteq G\}$. It is monotone: $G \subseteq G' \Rightarrow F(G) \subseteq F(G')$ —more structure never hides a channel.

Observing through $F(G)$ deposits the Fisher information (4) of exactly those channels, and that information is what licenses the next expansion (§3): a structural element $\omega \in \Omega$ becomes *data-supported* only when the channels loading on it are unlocked. Collect one round of supported growth into an operator.

Definition 2 (Unlock operator). $\Phi(G) = G \cup \{\omega \in \Omega : \omega \text{ is data-supported under the channels } F(G)\}$. It is *extensive* ($G \subseteq \Phi(G)$) and *monotone* ($G \subseteq G' \Rightarrow \Phi(G) \subseteq \Phi(G')$), because a wider frontier can only add support.

A community’s trajectory is the orbit $G_c(t+1) = \Phi(G_c(t))$ from its seed: structure unlocks channels, channels supply Fisher, Fisher supports expansion, expansion is more structure (Fig. 5a). Because Ω is finite the orbit is an increasing chain in 2^Ω and halts.

Proposition 1 (Seed-determined closure and structural pluralism). *On the finite lattice 2^Ω , Φ monotone and extensive has, above any seed G_0 , a unique least fixed point $G^*(G_0) = \bigcup_{t \geq 0} \Phi^t(G_0)$*

(Knaster–Tarski; Tarski, 1955). The learned structure is a function of the seed alone. Hence two disconnected communities with seeds G_0^A, G_0^B for which $G^*(G_0^A) \neq G^*(G_0^B)$ end with different unlocked frontiers $F(G^*)$: they cannot be brought to the same evidence, because each lacks a channel the other’s theory made visible. Pluralism is the multiplicity of fixed points of the unlock operator, not a tuned exploration term.

The family of structures reachable from a seed—closed under the order in which supported elements are taken—is an *accessible set system*; when prerequisites compose monotonically it is an *antimatroid*, the order-theoretic skeleton of a dependency “tech tree” (Korte et al., 1991; Davey & Priestley, 2002), with Φ its closure operator in the sense of formal concept analysis (Ganter & Wille, 1999). The reading is exactly curriculum learning with an endogenous curriculum (Bengio et al., 2009): what a community can next learn to see is set by what it has already learned. It is also Stanford’s problem of unconceived alternatives made structural—a channel never unlocked is not disbelieved but *unconceived*, and no amount of the evidence it would carry can reach the community (Stanford, 2006). Because the world is a rugged landscape (§2) with no single best basin for a bounded learner (Kauffman, 1993; Wolpert & Macready, 1997), different seeds settling in different basins is the generic case, not a pathology.

What couples the communities is the social graph. Let G_{soc} be a stochastic block model over communities—`graphs.community(sizes, intra, inter)` in the repository—with connectivity measured by the Fiedler value $\lambda_2 = \text{algebraic_connectivity}(G_{\text{soc}})$. Communities that share data-supported proposals advance Φ on the *union* of their held structures, merging their frontiers. Thus λ_2 is the lever on Proposition 1 (Fig. 5b): at $\lambda_2 = 0$ the blocks are separate echo chambers and seed-distinct fixed points persist (pluralism preserved); raising λ_2 merges frontiers and drives the population toward a common fixed point (consensus, or shared lock-in). Connection collapses structural divergence; disconnection protects it.

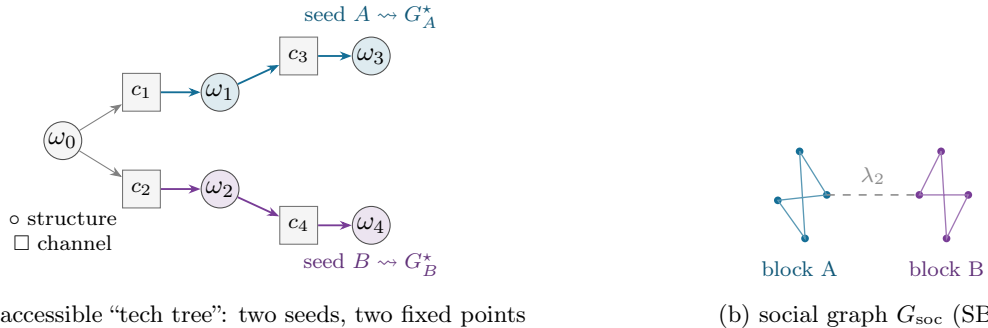


Figure 5: The endogenous frontier. a Held structures (o) unlock channels ($\square$) whose data support the next structure; a seed determines a route and a fixed point, and two seeds (teal A, purple B) reach *different* learned structures on the *same* world. b The Fiedler value λ_2 of the social block model sets whether frontiers stay separate ($\lambda_2=0$, pluralism) or merge ($\lambda_2>0$, consensus).

6 Synthesis: one object, four readings

The four sections are one environment seen from four fields (Table 1). What is true is an *energy landscape*; how it is represented and learned is a *graphical model under structure search*; where a community starts is a *coarse-graining*; and what it can come to see is an *accessible poset* whose fixed points are its possible paradigms. The composition is the whole claim: a bounded learner, seeded with a coarse model, climbs the partition lattice by expansion only through the channels its current structure unlocks, and so settles—path-dependently—into one basin of a landscape that admits

many. Nothing in the loop is a free exploration parameter; the diversity that protects a population is the multiplicity of fixed points, and the one social dial, λ_2 , decides whether that diversity survives.

Reading	Object	Borrowed from	Closed form / handle
Energy landscape	world law $p(x) \propto e^{-E(x)}$	Gibbs / spin glass (Hopfield, 1982; Mézard et al., 1987)	$\Pi^* = \text{energy Hessian}$ (2)
Graphical model	belief (Π_c, h_c) , learning	PGM + active inference (Pearl, 2009; Friston et al., 2018)	Fisher J_k , BMR ΔF (4),(5)
Coarse-graining	seed P_c , super-nodes	RG / lumpability (Wilson, 1975; Kemeny & Snell, 1976)	$\Pi_c = (S_c \Sigma^* S_c^\top)^{-1}$ (6)
Accessible poset	frontier F , closure Φ	order theory (Korte et al., 1991; Tarski, 1955)	least fixed point G^* (Prop. 1)

Table 1: The dictionary. Each row is the same environment under a different field’s first principles; the right column is the quantity that makes the reading computable.

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